

Maxima and Minima of a function of Two or More Variables

➤ Relative Maximum or local maximum and Relative Minimum or local minimum

Let $f(x, y)$ be a real valued function, and let (a, b) be a point in the domain of f . We say that f has a local maximum at (a, b) if $f(x, y) \leq f(a, b)$ for all x, y inside some disk of positive radius centered at (a, b) .

Likewise, we say that f has a local minimum at (a, b) if $f(x, y) \geq f(a, b)$ for all x, y inside some disk of positive radius centered at (a, b) .

If $f(x, y) \leq f(a, b)$ for all (x, y) in the domain of f , then f has a global maximum at (a, b) .

If $f(x, y) \geq f(a, b)$ for all (x, y) in the domain of f , then f has a global minimum at (a, b) .

OR

A function $f(x, y)$ is said to have a relative maximum value or local maximum value at the point (a, b) if

$$f(a + h, b + k) \leq f(a, b)$$

whatever be the relative values of the increments h, k positive or negative, provided they be taken sufficiently small and be finite and the point (a, b) is called local maximum point.

A function $f(x, y)$ is said to have a relative minimum value or local minimum value at the point (a, b) if

$$f(a + h, b + k) \geq f(a, b)$$

whatever be the relative values of the increments h, k positive or negative, provided they be taken sufficiently small and be finite and the point (a, b) is called local minimum point.

Extreme Value:

A maximum or a minimum value of a function is called an extreme value.

Critical Point or Stationary Point:

The point (a, b) is called the critical point of the function of two variables $f(x, y)$ if $f_x(a, b) = 0$ and $f_y(a, b) = 0$. In vector form, $\nabla f(a, b) = 0$.

Saddle Point:

The point $(a, b, f(a, b))$ is a saddle point of $z = f(x, y)$ if (a, b) is a critical point of f and if every open disk centered at (a, b) contains points (x, y) in the domain of f for which $f(x, y) < f(a, b)$ and points (x, y) in the domain of f for which $f(x, y) > f(a, b)$.

OR, The point at a critical point (a, b) in which the function of two variables is neither maximum nor minimum is called saddle point.

- State and prove the necessary condition for existence of extreme values of functions of several variables.

Statement:

If the function $f(x, y, z, \dots)$ is a function of n independent variables $x, y, z, \dots$, then the maximum and minimum values of f will exist if the first partial derivatives exist and

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = \frac{\partial f}{\partial z} = \dots = 0.$$

Proof:

Let us consider that function $f(x, y, z, \dots)$ is a function of n independent variables $x, y, z, \dots$, with the assumption that f is continuous and all order partial derivable and finite for all values of $x, y, z, \dots$ in the neighbourhood of $x = a, y = b, z = c, \dots$ i.e., of $(a, b, c, \dots)$. Then by Taylor's theorem we get

$$\begin{aligned} f(a + h, b + k, c + l, \dots) &= f(a, b, c, \dots) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} + l \frac{\partial}{\partial z} + \dots \right) f(x, y, z, \dots) \Big|_{(a, b, c, \dots)} \\ &+ \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} + l \frac{\partial}{\partial z} + \dots \right)^2 f(x, y, z, \dots) \Big|_{(a, b, c, \dots)} + \dots \end{aligned}$$

Here h, k, l are very small so that by neglecting 2nd and higher degree term we get, (here $0 < h, k, l, \dots < 1$)

$$\begin{aligned} f(a + h, b + k, c + l, \dots) &= f(a, b, c, \dots) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} + l \frac{\partial}{\partial z} + \dots \right) f(x, y, z, \dots) \Big|_{(a, b, c, \dots)} \\ \Rightarrow f(a + h, b + k, c + l, \dots) - f(a, b, c, \dots) &= h \left(\frac{\partial f}{\partial x} \right)_{(a, b, c, \dots)} + k \left(\frac{\partial f}{\partial y} \right)_{(a, b, c, \dots)} + l \left(\frac{\partial f}{\partial z} \right)_{(a, b, c, \dots)} + \dots \end{aligned} \quad \dots \dots \dots (1)$$

Now if $f(a, b, c, \dots)$ is maximum value, then

$$\begin{aligned} f(a + h, b + k, c + l, \dots) &\leq f(a, b, c, \dots) \\ \Rightarrow f(a + h, b + k, c + l, \dots) - f(a, b, c, \dots) &\leq 0 \end{aligned}$$

$$\text{i.e., } h \left(\frac{\partial f}{\partial x} \right)_{(a, b, c, \dots)} + k \left(\frac{\partial f}{\partial y} \right)_{(a, b, c, \dots)} + l \left(\frac{\partial f}{\partial z} \right)_{(a, b, c, \dots)} + \dots \leq 0 \quad \dots \dots \dots (2)$$

Again if $f(a, b, c, \dots)$ is minimum value, then

$$f(a + h, b + k, c + l, \dots) \geq f(a, b, c, \dots)$$

$$\Rightarrow f(a + h, b + k, c + l, \dots) - f(a, b, c, \dots) \geq 0$$

$$\text{i.e., } h \left(\frac{\partial f}{\partial x} \right)_{(a,b,c,\dots)} + k \left(\frac{\partial f}{\partial y} \right)_{(a,b,c,\dots)} + l \left(\frac{\partial f}{\partial z} \right)_{(a,b,c,\dots)} + \dots \geq 0 \quad \dots\dots\dots(3)$$

Now from (1) and (2), we have

$$h \left(\frac{\partial f}{\partial x} \right)_{(a,b,c,\dots)} + k \left(\frac{\partial f}{\partial y} \right)_{(a,b,c,\dots)} + l \left(\frac{\partial f}{\partial z} \right)_{(a,b,c,\dots)} + \dots = 0$$

$$\Rightarrow \left(\frac{\partial f}{\partial x} \right)_{(a,b,c,\dots)} = \left(\frac{\partial f}{\partial y} \right)_{(a,b,c,\dots)} = \left(\frac{\partial f}{\partial z} \right)_{(a,b,c,\dots)} = \dots = 0 \quad [\text{as } h, k, l, \dots \neq 0]$$

i.e., at the point $(a, b, c, \dots)$,

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = \frac{\partial f}{\partial z} = \dots = 0.$$

Smooth Function:

A function whose partial derivatives of all orders exist and are continuous is called smooth function or C^∞ function.

A real valued function whose first derivative is continuous is called continuously differentiable or a C^1 function.

- State and prove the sufficient conditions for existence of extreme values of functions of two variables.

Statement:

Let $f(x, y)$ be a smooth real-valued function, with a critical point at (a, b) , i.e., $f_x(a, b) = 0$, $f_y(a, b) = 0$ and let $r = f_{xx}(a, b)$, $s = f_{xy}(a, b)$, $t = f_{yy}(a, b)$, then

- (a) If $rt - s^2 > 0$ and $r < 0, t < 0$, then f has a local maximum at (a, b) .
- (b) If $rt - s^2 > 0$ and $r > 0, t > 0$, then f has a local minimum at (a, b) .
- (c) If $rt - s^2 < 0$, then f has neither a local maximum nor a local minimum at (a, b) i.e., f has a saddle point at (a, b) .
- (d) If $rt - s^2 = 0$, then no conclusion can be drawn, and we need further investigation.

Proof:

From the Taylor's expansion of the function of two variables, we have

$$\begin{aligned}
f(a+h, b+k) &= f(a, b) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right) f(x, y)|_{(a,b)} + \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^2 f(x, y)|_{(a,b)} \\
&\quad + R_3
\end{aligned}$$

where R_3 denotes the third degree of h, k and higher degree terms.

$$\begin{aligned}
&= f(a, b) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right) f(a, b) + \frac{1}{2!} \left(h^2 \frac{\partial^2}{\partial x^2} + 2hk \frac{\partial^2}{\partial x \partial y} + k^2 \frac{\partial^2}{\partial y^2}\right) f(a, b) + R_3 \\
&= f(a, b) + hf_x(a, b) + kf_y(a, b) + \frac{1}{2!} \{h^2 f_{xx}(a, b) + 2hk f_{xy}(a, b) + k^2 f_{yy}(a, b)\} + R_3 \\
&\quad \dots\dots\dots(1)
\end{aligned}$$

Since $f_x(a, b) = 0$, $f_{xy}(a, b) = 0$ and let $r = f_{xx}(a, b)$, $s = f_{xy}(a, b)$, $t = f_{yy}(a, b)$ so from (1) we have

$$f(a+h, b+k) = f(a, b) + h \cdot 0 + k \cdot 0 + \frac{1}{2!} (h^2 r + 2hks + k^2 t) + R_3$$

Neglecting the higher power of h, k which are very small ($0 < h, k < 1$), we get

$$\begin{aligned}
f(a+h, b+k) - f(a, b) &= \frac{1}{2!r} (h^2 r^2 + 2hkrs + k^2 rt) \\
&= \frac{1}{2!r} \{(hr + sk)^2 + rtk^2 - s^2 k^2\} \\
&= \frac{1}{2!r} \{(hr + sk)^2 + (rt - s^2)k^2\} \quad \dots\dots\dots (2)
\end{aligned}$$

Here $f(a, b)$ will be maximum if

$$\begin{aligned}
&f(a+h, b+k) - f(a, b) < 0 \\
&= \frac{1}{2!r} \{(hr + sk)^2 + (rt - s^2)k^2\} < 0 \quad [\text{by (2)}] \\
&\Rightarrow \text{if } r < 0 \quad \text{and} \quad rt - s^2 > 0 \\
&\Rightarrow \text{if } r < 0 \quad \text{and} \quad rt - s^2 > 0, \text{ again } rt - s^2 > 0 \text{ if } t < 0
\end{aligned}$$

Again, (a, b) will be minimum if

$$\begin{aligned}
&f(a+h, b+k) - f(a, b) > 0 \\
&= \frac{1}{2!r} \{(hr + sk)^2 + (rt - s^2)k^2\} > 0 \quad [\text{by (2)}] \\
&\Rightarrow \text{if } r > 0 \quad \text{and} \quad rt - s^2 > 0
\end{aligned}$$

$\Rightarrow$ if $r > 0$ and $rt - s^2 > 0$, again $rt - s^2 > 0$ if $t > 0$

$\Rightarrow$ if $rt - s^2 > 0$ and $r > 0$, $t > 0$.

If $rt - s^2 < 0$, then $f(a + h, b + k) - f(a, b) > 0$ or < 0 . Hence (a, b) is a saddle point i.e., $f(x, y)$ has neither a maximum or a minimum at (a, b) .

And if $rt - s^2 = 0$ then the case is doubtful; and needs further investigation.

- State and prove the sufficient conditions for determining extreme values of functions of three variables.

Statement:

Let $f(x, y, z)$ be a real-valued smooth function with a critical point at (a, b, c) .

Then $f_x(a, b, c) = 0$, $f_y(a, b, c) = 0$, $f_z(a, b, c) = 0$. Let $A = f_{xx}(a, b, c)$, $B = f_{yy}(a, b, c)$, $C = f_{zz}(a, b, c)$, $F = f_{yz}(a, b, c)$, $G = f_{zx}(a, b, c)$, $H = f_{xy}(a, b, c)$. Then

- $f(x, y, z)$ will have maximum value at (a, b, c) if $A < 0$, $\begin{vmatrix} A & H \\ H & B \end{vmatrix} > 0$, $\begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix} < 0$.
- $f(x, y, z)$ will have minimum value at (a, b, c) if $A > 0$, $\begin{vmatrix} A & H \\ H & B \end{vmatrix} > 0$, $\begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix} > 0$.
- $f(x, y, z)$ will have neither maximum nor minimum value at (a, b, c) if the above two conditions do not hold.

Proof:

From the Taylor's expansion of the function of two variables, we have

$$f(a + h, b + k, c + l) = f(a, b, c) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} + l \frac{\partial}{\partial z} \right) f(a, b, c) + \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} + l \frac{\partial}{\partial z} \right)^2 f(a, b, c) + R_3$$

where R_3 denotes the third degree of h, k and higher degree terms.

$$= f(a, b, c) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} + l \frac{\partial}{\partial z} \right) f(a, b, c) \\ + \frac{1}{2!} \left(h^2 \frac{\partial^2}{\partial x^2} + k^2 \frac{\partial^2}{\partial y^2} + l^2 \frac{\partial^2}{\partial z^2} + 2hk \frac{\partial^2}{\partial x \partial y} + 2kl \frac{\partial^2}{\partial y \partial z} + 2lh \frac{\partial^2}{\partial z \partial x} \right) f(a, b, c) + R_3$$

$$\Rightarrow f(a + h, b + k, c + l) - f(a, b, c) = hf_x + kf_y + lf_z \\ + \frac{1}{2!} (hf_{xx} + kf_{yy} + lf_{zz} + 2hkf_{xy} + 2klf_{yz} + 2lhf_{zx}) + R_3 \quad \dots\dots\dots (i)$$

Since the necessary condition of extreme values is

$$f_x = 0, f_y = 0, f_z = 0, \text{ so from (i) we get}$$

$$f(a + h, b + k, c + l) - f(a, b, c) = \frac{1}{2!} (hA + kB + lC + 2hkH + 2klF + 2lhG) + R_3$$

Since h, k, l are sufficiently small, so neglecting the third and higher degree terms we get

$$f(a + h, b + k, c + l) - f(a, b, c) = \frac{1}{2!} (hA + kB + lC + 2hkH + 2klF + 2lhG) \\ = \frac{1}{2!A} (hA^2 + kAB + lAC + 2hkAH + 2klAF + 2lhAG) \\ = \frac{1}{2A} [(Ah + Gl + Hk)^2 - G^2l^2 - H^2k^2 - 2GHkl + ABk^2 + ACl^2 + 2AFkl] \\ = \frac{1}{2A} [(Ah + Gl + Hk)^2 + (AB - H^2)k^2 + 2(AF - GH)kl + (AC - G^2)l^2]$$

For maximum value, we have

$$f(a + h, b + k, c + l) - f(a, b, c) < 0 \\ \Rightarrow \frac{1}{2A} [(Ah + Gl + Hk)^2 + (AB - H^2)k^2 + 2(AF - GH)kl + (AC - G^2)l^2] < 0 \\ \Rightarrow A < 0 \text{ and } (Ah + Gl + Hk)^2 + (AB - H^2)k^2 + 2(AF - GH)kl + (AC - G^2)l^2 > 0 \\ \Rightarrow A < 0 \text{ and } (AB - H^2)k^2 + 2(AF - GH)kl + (AC - G^2)l^2 > 0 \\ \Rightarrow A < 0, AB - H^2 > 0 \text{ and } (AB - H^2)(AC - G^2) - (AF - GH)^2 > 0$$

$$\left[ax^2 + 2hxy + by^2 > 0 \text{ if } a > 0 \text{ and } ab - h^2 > 0 \right]$$

$$\Rightarrow A < 0, AB - H^2 > 0 \text{ and } A(ABC + 2FGH - AF^2 - BG^2 - CH^2) > 0$$

$$\Rightarrow A < 0, AB - H^2 > 0 \text{ and } ABC + 2FGH - AF^2 - BG^2 - CH^2 < 0$$

$$\Rightarrow A < 0, \begin{vmatrix} A & H \\ H & B \end{vmatrix} > 0 \text{ and } \begin{vmatrix} A & H & F \\ H & B & F \\ G & F & C \end{vmatrix} > 0$$

which is the required condition of $f(x, y, z)$ to have maximum value at (a, b, c) .

For minimum value, we have

$$f(a + h, b + k, c + l) - f(a, b, c) > 0$$

$$\Rightarrow \frac{1}{2A} \left[(Ah + Gl + Hk)^2 + (AB - H^2)k^2 + 2(AF - GH)kl + (AC - G^2)l^2 \right] > 0$$

$$\Rightarrow A > 0 \text{ and } (Ah + Gl + Hk)^2 + (AB - H^2)k^2 + 2(AF - GH)kl + (AC - G^2)l^2 > 0$$

$$\Rightarrow A > 0 \text{ and } (AB - H^2)k^2 + 2(AF - GH)kl + (AC - G^2)l^2 > 0$$

$$\Rightarrow A > 0, AB - H^2 > 0 \text{ and } (AB - H^2)(AC - G^2) - (AF - GH)^2 > 0$$

$$\left[ax^2 + 2hxy + by^2 > 0 \text{ if } a > 0 \text{ and } ab - h^2 > 0 \right]$$

$$\Rightarrow A > 0, AB - H^2 > 0 \text{ and } A(ABC + 2FGH - AF^2 - BG^2 - CH^2) > 0$$

$$\Rightarrow A > 0, AB - H^2 > 0 \text{ and } ABC + 2FGH - AF^2 - BG^2 - CH^2 < 0$$

$$\Rightarrow A > 0, \begin{vmatrix} A & H \\ H & B \end{vmatrix} > 0 \text{ and } \begin{vmatrix} A & H & F \\ H & B & F \\ G & F & C \end{vmatrix} > 0$$

which is the required condition of $f(x, y, z)$ to have minimum value at (a, b, c) .

where, $A = f_{xx}, B = f_{yy}, C = f_{zz}, F = f_{yz}, G = f_{zx}, H = f_{xy}$.

Problem – 01:

Find the maximum and minimum values of the function

$$f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4.$$

Solution:

The given function is

$$f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$$

$$\therefore f_x = 3x^2 + 3y^2 - 6x \text{ and } f_y = 6xy - 6y$$

$$\therefore r = f_{xx} = 6x - 6, s = f_{xy} = 6y \text{ and } t = f_{yy} = 6x - 6$$

For the existence of maximum and minimum values of the given function we have

$$f_x = 0 \quad \text{and} \quad f_y = 0$$

$$\therefore 3x^2 + 3y^2 - 6x = 0 \quad \text{and} \quad 6xy - 6y = 0$$

$$\text{i.e. } x^2 + y^2 - 2x = 0 \dots\dots\dots (ii) \quad \text{and} \quad x(y-1) = 0 \text{ or } y = 0, x = 1$$

When $y = 0$ then from (ii), we get $x = 0, 2$. So the points are $(0,0), (2,0)$.

Again when $x = 1$, then from (ii) we get $y = \pm 1$. So the points are $(1,1), (1,-1)$.

Hence the critical points are $(0,0), (2,0), (1,1), (1,-1)$.

At the critical point $(0,0)$, we have $r = -6 < 0, s = 0, t = -6 < 0$ and $rt - s = 36 > 0$.

So the function has maximum value at the point $(0,0)$ and the maximum value is

$$f_{\max} = 0^3 + 3.0.0^2 - 3(0)^2 - 3(0)^2 + 4 = 4$$

At the critical point $(2,0)$, we have $r = 6 > 0, s = 0, t = 6 > 0$ and $rt - s = 36 > 0$.

So the function has minimum value at the point $(0,0)$ and the minimum value is

$$f_{\min} = 2^3 + 3.2.0^2 - 3(2)^2 - 3(0)^2 + 4 = 8 - 12 + 4 = 0$$

At the critical point $(1,1)$, we have $r = 0, s = 6, t = 0$ and $rt - s = -36 < 0$.

So the function has neither maximum nor minimum value at the point $(1,1)$ and the point is saddle point.

Again, at the critical point $(1,-1)$, we have $r = 0, s = -6, t = 0$ and $rt - s = -36 < 0$.

So the function has neither maximum nor minimum value at the point $(1,-1)$ and the point is saddle point.

Problem – 02:

Find the relative maximum and minimum values of the function

$$f(x, y) = x^3 + y^3 - 3x - 12y + 20.$$

Solution:

The given function is

$$f(x, y) = x^3 + y^3 - 3x - 12y + 20$$

$$\therefore f_x = 3x^2 - 3 \text{ and } f_y = 3y^2 - 12$$

$$\therefore r = f_{xx} = 6x, s = f_{xy} = 0 \text{ and } t = f_{yy} = 6y$$

For the existence of maximum and minimum values of the given function we have

$$f_x = 0 \quad \text{and} \quad f_y = 0$$

$$\therefore 3x^2 - 3 = 0 \quad \text{and} \quad 3y^2 - 12 = 0$$

$$\text{i.e. } x = \pm 1 \quad \text{and} \quad y = \pm 2$$

So the critical points are $(1, 2), (1, -2), (-1, 2), (-1, -2)$.

At the critical point $(1, 2)$, we have $r = 6 > 0, s = 0, t = 12 > 0$ and $rt - s = 72 > 0$.

So the function has minimum value at the point $(0, 0)$ and the minimum value is

$$f_{\min} = 1 + 8 - 3 - 24 + 20 = 2$$

At the critical point $(1, -2)$, we have $r = 6 > 0, s = 0, t = -12 < 0$ and $rt - s = -72 < 0$.

So the function has neither maximum nor minimum value at the point $(1, -2)$ and the point is saddle point.

At the critical point $(-1, 2)$, we have $r = -6 < 0, s = 0, t = 12 > 0$ and $rt - s = -72 < 0$.

So the function has neither maximum nor minimum value at the point $(-1, 2)$ and the point is saddle point.

So the function has minimum value at the point $(0, 0)$ and the minimum value is

$$f_{\min} = 2^3 + 3.2.0^2 - 3(2)^2 - 3(0)^2 + 4 = 8 - 12 + 4 = 0$$

At the critical point $(1, 1)$, we have $r = 0, s = 6, t = 0$ and $rt - s = -36 < 0$.

Again, at the critical point $(-1, -2)$, we have $r = -6 < 0, s = 0, t = -12 < 0$ and $rt - s = 72 > 0$.

So the function has neither maximum nor minimum value at the point $(1, -1)$ and the point is saddle point.

So the function has maximum value at the point $(-1, -2)$ and the maximum value is

$$f_{\max} = -1 - 8 + 3 + 24 + 20 = 38$$

Problem – 03:

Find the maximum and minimum values of the function

$$f(x, y) = x^3 y^2 (1 - x - y).$$

Solution:

The given function is

$$f(x, y) = x^3 y^2 (1 - x - y)$$

$$\text{i.e. } f(x, y) = x^3 y^2 - x^4 y^2 - x^3 y^3$$

$$\therefore f_x = 3x^2 y^2 - 4x^3 y^2 - 3x^2 y^3 \text{ and } f_y = 2x^3 y - 2x^4 y - 3x^3 y^2$$

$$\therefore r = f_{xx} = 6xy^2 - 12x^2 y^2 - 6xy^3, s = f_{xy} = 6x^2 y - 8x^3 y - 9x^2 y^2 \text{ and}$$

$$t = f_{yy} = 2x^3 - 2x^4 - 6x^3 y$$

For the existence of maximum and minimum values of the given function we have

$$f_x = 0 \quad \text{and} \quad f_y = 0$$

$$\therefore 3x^2 y^2 - 4x^3 y^2 - 3x^2 y^3 = 0 \quad \text{and} \quad 2x^3 y - 2x^4 y - 3x^3 y^2 = 0$$

$$\text{i.e. } 3 - 4x - 3y = 0 \quad \dots\dots\dots (i) \quad \text{and} \quad 2 - 2x - 3y = 0 \quad \dots\dots\dots (ii)$$

Now subtracting (i) from (ii), we get

$$3 - 4x - 3y - 2 - 2x + 3y = 0 \quad \text{i.e. } x = \frac{1}{2}$$

Now putting the value of x in (ii), we get $x = \frac{1}{3}$

So the critical point of the given function is $\left(\frac{1}{2}, \frac{1}{3}\right)$.

At the critical point $\left(\frac{1}{2}, \frac{1}{3}\right)$:

$$\text{We have } r = 6 \cdot \frac{1}{2} \cdot \frac{1}{9} - 12 \cdot \frac{1}{4} \cdot \frac{1}{9} - 6 \cdot \frac{1}{2} \cdot \frac{1}{27} = -\frac{1}{9} > 0$$

$$s = 6 \cdot \frac{1}{2} \cdot \frac{1}{9} - 8 \cdot \frac{1}{8} \cdot \frac{1}{3} - 9 \cdot \frac{1}{4} \cdot \frac{1}{9} = -\frac{1}{12}$$

$$t = 2 \cdot \frac{1}{8} - 2 \cdot \frac{1}{16} - 6 \cdot \frac{1}{8} \cdot \frac{1}{3} = -\frac{1}{8} < 0$$

$$\text{and } rt - s = \frac{1}{72} - \frac{1}{144} = \frac{1}{144} > 0.$$

So the function has maximum value at the point $\left(\frac{1}{2}, \frac{1}{3}\right)$ and the maximum value is

$$f_{\max} = \frac{1}{8} \cdot \frac{1}{9} \left(1 - \frac{1}{2} - \frac{1}{3}\right) = \frac{1}{432}$$

Problem – 04:

Find the maximum and minimum values of the function

$$f(x, y) = x^2y^2 - 5x^2 - 8xy - 5y^2.$$

Solution:

The given function is

$$f(x, y) = x^2y^2 - 5x^2 - 8xy - 5y^2$$

$$\therefore f_x = 2xy^2 - 10x - 8y \text{ and } f_y = 2x^2y - 8x - 10y$$

$$\therefore r = f_{xx} = 2y^2 - 10, s = f_{xy} = 4xy - 8 \text{ and } t = f_{yy} = 2x^2 - 10$$

For the existence of maximum and minimum values of the given function we have

$$f_x = 0 \quad \text{and} \quad f_y = 0$$

$$\therefore 2xy^2 - 10x - 8y = 0 \quad \text{and} \quad 2x^2y - 8x - 10y = 0$$

$$\text{i.e. } xy^2 - 5x - 4y = 0 \dots\dots\dots (i) \quad \text{and} \quad x^2y - 4x - 5y = 0 \dots\dots\dots (ii)$$

Now subtracting (i) from (ii), we get

$$xy^2 - 5x - 4y - x^2y + 4x + 5y = 0$$

$$\Rightarrow xy(y - x) - 4(y - x) + 5(y - x) = 0$$

$$\Rightarrow (y - x)(xy + 1) = 0$$

$$\Rightarrow x = y \dots\dots\dots (iii) \text{ and } x = -\frac{1}{y} \dots\dots\dots (iv)$$

Now putting $x = y$ in (i), we get

$$y^3 - 9y = 0 \therefore y = 0, \pm 3.$$

So from (iii), we get

$$\text{when } y = 0, \text{ then } x = 0, \pm 3$$

$$\text{and when } y = \pm 3, \text{ then } x = \pm 3$$

Again, putting $\Rightarrow x = -\frac{1}{y}$ in (i), we get

$$-y + \frac{5}{y} - 4y = 0 \text{ or } \frac{5}{y} - 5y = 0 \text{ or } 1 - y^2 = 0 \therefore y = \pm 1$$

So from (iii), we get

$$\text{when } y = 1, \text{ then } x = -1$$

$$\text{and when } y = -1, \text{ then } x = 1$$

So the critical points are $(0,0), (3,3), (-3,-3), (-1,1), (1,-1)$.

At the critical point $(0,0)$, we have $r = -10 < 0, s = -8, t = -10 < 0$ and $rt - s = 36 > 0$.

So the function has maximum value at the point $(0,0)$ and the maximum value is

$$f_{\max} = 0$$

At the critical point $(3,3)$, we have $r = 8 > 0, s = 28, t = 8 > 0$ and $rt - s = 64 - (28)^2 < 0$.

So the function has neither maximum nor minimum value at the point $(1,-2)$ and the point is saddle point.

At the critical point $(-3,-3)$, we have $r = 8 > 0, s = 28, t = 8 > 0$ and $rt - s = 64 - (28)^2 < 0$.

So the function has neither maximum nor minimum value at the point $(-1,2)$ and the point is saddle point.

At the critical point $(-1,1)$, we have $r = -8 < 0, s = -12, t = -8 < 0$ and $rt - s = 64 - (-12)^2 < 0$.

So the function has neither maximum nor minimum value at the point $(-1,1)$ and the point is saddle point.

Again, at the critical point $(1,-1)$, we have $r = -8 < 0, s = -12, t = -8 < 0$ and $rt - s = 64 - (-12)^2 < 0$.

So the function has neither maximum nor minimum value at the point $(-1,1)$ and the point is saddle point.

Hence the points $(3,3), (-3,-3), (-1,1)$ and $(1,-1)$ are saddle points.

Problem – 05:

Examine the maximum and minimum values of the function

$$f(x, y) = \sin x + \sin y + \sin(x + y).$$

Solution:

The given function is

$$f(x, y) = \sin x + \sin y + \sin(x + y) \dots\dots\dots (i)$$

$$\therefore f_x = \cos x + \cos(x + y) \text{ and } f_y = \cos y + \cos(x + y)$$

$$\therefore r = f_{xx} = -\sin x - \sin(x + y), s = f_{xy} = -\sin(x + y) \text{ and } t = f_{yy} = -\sin y - \sin(x + y)$$

For the existence of maximum and minimum values of the given function we have

$$f_x = 0 \text{ and } f_y = 0$$

$$\text{i.e. } \cos x + \cos(x + y) = 0 \dots\dots\dots (ii) \text{ and } \cos y + \cos(x + y) = 0 \dots\dots\dots (iii)$$

Now subtracting (iii) from (ii), we get

$$\cos x - \cos y = 0 \text{ or } x = y \dots\dots\dots (iv)$$

Putting $x = y$ in (ii) we have

$$\cos x + \cos(x + x) = 0$$

$$\Rightarrow \cos x + \cos 2x = 0$$

$$\Rightarrow \cos 2x = -\cos x$$

$$\Rightarrow \cos 2x = \cos(\pi - x)$$

$$\Rightarrow 2x = \pi - x$$

$$\Rightarrow x = \frac{\pi}{3}$$

From (iv), we find

$$y = \frac{\pi}{3}$$

So the critical point is $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$.

At the critical point $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$:

$$\text{We have } r = -\sin \frac{\pi}{3} - \sin \frac{2\pi}{3} = -\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} = -\sqrt{3} < 0$$

$$s = -\sin \frac{2\pi}{3} = -\frac{\sqrt{3}}{2}$$

$$t = -\sin \frac{\pi}{3} - \sin \frac{2\pi}{3} = -\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} = -\sqrt{3} < 0$$

$$\text{and } rt - s^2 = 3 - \frac{3}{4} = \frac{9}{4} > 0.$$

So the function has maximum value at the point $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$ and the maximum value is

$$f_{\max} = \sin \frac{\pi}{3} + \sin \frac{\pi}{3} + \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}.$$

Problem – 01:

Find the maximum and minimum values of the function

$$f(x, y, z) = (x + y + z)^3 - 3(x + y + z) - 24xyz + a^2.$$

or ,

Prove that the function $f(x, y, z) = (x + y + z)^3 - 3(x + y + z) - 24xyz + a^2$ has minimum value at (1,1,1) and maximum value at (-1,-1,-1).

Solution:

The given function is

$$f(x, y, z) = (x + y + z)^3 - 3(x + y + z) - 24xyz + a^2$$

$$\therefore f_x = 3(x+y+z)^2 - 3 - 24yz, f_y = 3(x+y+z)^2 - 3 - 24zx, f_z = 3(x+y+z)^2 - 3 - 24xy$$

$$\text{Let } A = f_{xx} = 6(x+y+z), B = f_{yy} = 6(x+y+z), C = f_{zz} = 6(x+y+z),$$

$$F = f_{yz} = 6(x+y+z) - 24x, G = f_{zx} = 6(x+y+z) - 24y, H = f_{xy} = 6(x+y+z) - 24z.$$

For the maximum and minimum values, we have

$$f_x = 0, f_y = 0, f_z = 0$$

$$\text{i.e. } 3(x+y+z)^2 - 3 - 24yz = 0 \quad \therefore (x+y+z)^2 - 1 - 8yz = 0 \quad \dots\dots\dots (i)$$

$$3(x+y+z)^2 - 3 - 24zx = 0 \quad \therefore (x+y+z)^2 - 1 - 8zx = 0 \quad \dots\dots\dots (ii)$$

$$3(x+y+z)^2 - 3 - 24xy = 0 \quad \therefore (x+y+z)^2 - 1 - 8xy = 0 \quad \dots\dots\dots (iii)$$

Solving (i), (ii) and (iii), we get

$$xy = yz = zx$$

$$\Rightarrow x = y = z$$

Now from (ii) we find

$$9x^2 - 1 - 8x^2 = 0 \quad \therefore x = \pm 1.$$

So the critical points are (1,1,1) and (-1,-1,-1).

At the point (1,1,1):

$$A = 18, B = 18, C = 18, F = -6, G = -6, H = -6$$

$$\text{Here, } A > 0, \begin{vmatrix} A & H \\ H & B \end{vmatrix} = \begin{vmatrix} 18 & -6 \\ -6 & 18 \end{vmatrix} = 288 > 0$$

$$\text{and } \begin{vmatrix} A & H & F \\ H & B & F \\ G & F & C \end{vmatrix} = \begin{vmatrix} 18 & -6 & -6 \\ -6 & 18 & -6 \\ -6 & -6 & 18 \end{vmatrix} = 3456 > 0$$

So at the point (1,1,1), the given function f has minimum value and

$$f_{\min} = 3^3 - 3.3 - 24 + a^2 = a^2 - 6.$$

At the point (-1,-1,-1):

$$A = -18, B = -18, C = -18, F = 6, G = 6, H = 6$$

Here, $A < 0$, $\begin{vmatrix} A & H \\ H & B \end{vmatrix} = \begin{vmatrix} -18 & 6 \\ 6 & -18 \end{vmatrix} = 288 > 0$

and $\begin{vmatrix} A & H & F \\ H & B & F \\ G & F & C \end{vmatrix} = \begin{vmatrix} -18 & 6 & 6 \\ 6 & -18 & 6 \\ 6 & 6 & -18 \end{vmatrix} = -3456 > 0$

So at the point $(-1, -1, -1)$, the given function f has minimum value and

$$f_{\max} = (-3)^3 - 3(-3) - 24 + a^2 = a^2 + 6$$

Constrained Optimization: Lagrange Multipliers

Theorem:

Suppose that $f(x, y, z)$ and $g(x, y, z)$ are functions with continuous first order partial derivatives and $\nabla g(x, y, z) \neq 0$ on the surface $g(x, y, z) = 0$. Suppose that either

- (1) The maximum value of $f(x, y, z)$ subject to the constraint $g(x, y, z) = 0$ occurs at (x_0, y_0, z_0) ; or
- (2) The minimum value of $f(x, y, z)$ subject to the constraint $g(x, y, z) = 0$ occurs at (x_0, y_0, z_0) ;

Then $\nabla f(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0)$ for some constant λ . This λ is called Lagrange multiplier.

Problem – 01:

Use Lagrange multipliers to find three real numbers whose sum is 12 and the sum of whose squares is minimum.

Solution:

Let the numbers be x, y, z . Given that $x + y + z = 12$. we have to minimize

$$f(x, y, z) = x^2 + y^2 + z^2 \dots\dots\dots (1)$$

$$\text{Let } g(x, y, z) = x + y + z - 12 = 0 \dots\dots\dots (2)$$

For extremum, we have

$$\begin{aligned} \nabla f(x, y, z) &= \lambda \nabla g(x, y, z) \\ \Rightarrow \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2) &= \lambda \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (x + y + z - 12) \end{aligned}$$

$$\Rightarrow (2x\underline{i} + 2y\underline{j} + 2z\underline{k}) = \lambda(\underline{i} + \underline{j} + \underline{k})$$

$$\Rightarrow 2x = 2y = 2z = \lambda$$

$$\Rightarrow x = y = z = \frac{\lambda}{2}$$

So from (2), we get $x = y = z = 4$